

CNS Infection

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- Infections of the nervous system can be caused by viruses, bacteria, fungi or protozoa.
 - They may affect the lining of the brain, CSF, brain parenchyma, spinal cord, nerve roots, peripheral nerve or muscle.
 - Infections of the CNS may be divided into meningitis, encephalitis, focal suppuration or inflammation.

Bacterial meningitis

- Bacterial meningitis is caused by a primary infection within the subarachnoid space that causes acute inflammation of the meninges (pia and arachnoid mater).
- The incidence of bacterial meningitis is 2–6/100,000 per annum with peaks in infancy, adolescence and the elderly.
- It may be caused by a variety of organisms which vary in frequency according to the age of the host and other risk factors.
- Bacterial infection reaches the CNS either by direct invasion, haematogenous spread or embolisation of infected thrombi.
- Also infection may be iatrogenic (e.g. following ventriculo-peritoneal shunt, intracranial pressure monitor or surgery).

Common causes of bacterial meningitis by age and risk factors

Neonates (<3 months)	Escherichia coli Group B streptococcus Listeria monocytogenes
Infant and child (>3 months)	Streptococcus pneumoniae (pneumococcus) Neisseria meningitidis (meningococcus) Haemophilus influenzae
Adults <50 (healthy and immunocompetent)	Streptococcus pneumoniae Neisseria meningitidis
Adult >50	Streptococcus pneumoniae Neisseria meningitidis Listeria monocytogenes



Skull fracture/post neurosurgery	Staphylococcus epidermidis, Staph. Aureus, Gram-negatives (Klebsiella, Proteus, Pseudomonas, E. coli, Serratia, Acinetobacter)
Pregnancy	Listeria monocytogenes
Immunosuppressed (e.g. malignancy, alcohol, diabetes mellitus, septicaemia, UTI)	Listeria monocytogenes Gram-negatives Streptococcus pneumoniae Pseudomonas aeruginosa Group B streptococcus Staph. aureus
Rarer causes	Salmonella, Shigella, Clostridium, Neisseria gonorrhoea

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- In community-acquired meningitis the transmission of pathogenic bacteria occurs by respiratory droplets spread or close contact with a carrier.
 - Bacteria initially colonize the nasopharynx by attaching to epithelial cells using outer adhesive pili and membrane proteins.
 - The risk of colonization of the respiratory epithelium is increased by damage caused by irritants such as a preceding viral illness or cigarette smoke.
 - Bacteria then enter the bloodstream and gain access to the CSF via the choroid plexus epithelium or cerebral capillaries.

Clinical presentation

- Meningitis is characterized by the presence of fever, headache and neck stiffness.
- There is usually associated nausea, vomiting, photophobia and progressive lethargy, stupor or coma, and epileptic activity may develop.
- Seizures may be focal or generalized and may occur either at presentation or at any time during the course of bacterial meningitis in up to 40% of patients.
- Meningitis may lead to vascular occlusion and subsequent infarction of the brain or spinal cord, progressive cranial nerve lesions and hydrocephalus secondary to impaired CSF absorption and obstruction.

Investigation

- Bacterial meningitis is accompanied by an elevated white cell count and raised inflammatory markers.
- In the presence of septicaemia, platelet consumption may occur with loss of clotting factors.
- There may be associated hypocalcaemia, hyponatraemia and impaired renal function with metabolic acidosis.
- Imaging should be undertaken urgently to exclude mass lesions, hydrocephalus or cerebral edema which are a contraindication to lumbar puncture.
- The diagnosis of meningitis is made by blood culture, CSF analysis, staining and culture.

Comparison of cerebrospinal fluid (CSF) findings in different forms of meningitis

	Normal	Acute bacterial meningitis	Viral meningitis	TB meningitis
Appearance	Clear colorless	Turbid	Clear/opalescent	Clear/opalescent
Pressure	Normal	Increased	Normal or increased	Increased
Cells	0–5/mm ³	100–60,000/mm ³	5–1000/mm ³	5–1000/mm ³
Polymorphs	None	>80%	<50%	<50%
Glucose	75% blood glucose	Low (<40% blood glucose)	Normal	Low <50% blood concentration
Protein	<0.4 g/L	>0.9 g/L (1–5 g/L)	>0.4–0.9 g/L	>1 g (1–5 g/L)

Treatment

- In patients with severe meningitis, the first priority is to commence empirical antibiotic treatment, usually immediately after sending a blood culture.
- Empirical therapy in the absence of staining and culture depends on age, immune status and any other risk factors.
- In healthy immunocompetent adults treatment should be initiated with a third generation cephalosporin and vancomycin.
- It is recommended to commence benzylpenicillin in any child suspected of having meningococcal meningitis.
- When there is the possibility of nosocomial infection (e.g. following surgery, trauma or in the presence of a shunt) ceftazidime is preferred because it is more active against *Pseudomonas* than other third generation cephalosporins.

Antibiotics used in empirical therapy of bacterial meningitis

Preterm infants to
infants <1 month
Infants 1–3 months

Ampicillin +
cefotaxime
Ampicillin +
cefotaxime or
ceftriaxone

Immunocompetent children
>3 months and adults <55 years

Cefotaxime or
ceftriaxone +
vancomycin

Adults >55 years and adults
of any age with alcoholism
or other debilitating illnesses

Ampicillin +
cefotaxime or
ceftriaxone +
vancomycin

Hospital-acquired meningitis,
posttraumatic or postneurosurgery
meningitis, neutropenic patients,
or patients with impaired
cell-mediated immunity

Ampicillin +
ceftazidime +
vancomycin

Antibiotics used in the therapy of bacterial meningitis based on pathogen

Neisseria meningitidis (meningococcus)	Ceftriaxone 2 g (IV) 12-hrly for 7 days or Cefotaxime 2 g (IV) 6-hrly for 7 days Alternatively: Benzylpenicillin (penicillin G) 2.4 g (IV) 4-hrly for 7 days or Chloramphenicol 12.5 mg/kg 6-hrly or Amoxicillin 2 g (IV) 4-hrly for 7 days
Streptococcus pneumoniae	Cefotaxime 2 g (i.v.) 4–6 hourly for 14 days and Vancomycin 1 g (i.v.) 12 hourly OR Ceftriaxone 2 g (i.v.) 12 hourly for 14 days and Vancomycin 1 g (i.v.) 12 hourly for 14 days
Haemophilus influenzae	Cefotaxime 2 g (i.v.) 4–6 hourly for 14 days OR Ceftriaxone 2 g (i.v.) 12 hourly for 14 days

Listeria monocytogenes	Ampicillin or Amoxillin 2 g (IV) 4-hrly for 3 weeks and gentamicin for 7 days Alternatively: Co-trimoxazole 3 mg/kg (IV) 6-hrly for 21 days.
Staphylococcus aureus	Vancomycin 1 g (i.v.) 12 hourly for 4 weeks
Pseudomonas	Ceftazidime 2 g (i.v.) 8 hourly for 3 weeks and gentamicin 3–5 mg/kg/day in divided doses, 8 hourly OR Meropenem 2 g (i.v.) 8 hourly for 4 weeks



Specific causes of bacterial meningitis

Neisseria meningitidis (meningococcal meningitis)

- This is a Gram-negative diplococcus. It is the most common identified cause of meningitis in children and young adults, with an incidence of 1–1.5/100,000.
- The majority of cases occur in the winter and early spring.
- Transmission occurs by droplet spread
- 5–10% of all adults are asymptomatic nasopharyngeal carriers of these pathogenic bacteria.
- The mortality is approximately 25% although this rises to 40% in those with meningococcal septicaemia coexisting with meningitis.

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- Clinical manifestations of meningococcal infection may develop within minutes or hours.
 - In approximately 40% of patients there is isolated meningitis; 10% have septicemia alone and the remainder a mixed pattern.
 - Isolated meningococcal meningitis carries a better prognosis than meningococcal septicemia or mixed picture.
 - In early meningococcal meningitis there is a diffuse erythematous macular papular rash which eventually develops into the characteristic petechiae found across the trunk and lower extremities, in the mucous membranes, conjunctiva and occasionally on the palms and soles.

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- The development of meningococcal septicemia is associated with progressive vasomotor disturbance culminating in profound hypotension, tachycardia and a rising respiratory rate indicating pulmonary edema or raised intracranial pressure resulting from cerebral edema.
 - **Waterhouse–Friderichsen** syndrome is a form of fulminant meningococcal disease, in which severe septicemia is complicated by the development of bilateral hemorrhage into the adrenal glands and disseminated intravascular coagulation leading to the development of severe sudden febrile illness associated with septic shock, petechiae, purpura and coma.

Meningococcal rash



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Meningococcal rash



Management

- Patients with probable fulminant meningococcal meningitis should be treated immediately in primary care with parenteral benzylpenicillin 2.4 g intravenously.
- They should be transferred immediately to A&E department.
- First line treatment is 2g ceftriaxone iv 12-hrly for 7 days.
- Alternatively cefotaxime 2g iv 6-hrly for 7 days or benzylpenicillin 2.4g iv 4-hrly for 7 days can be used
- Antibiotics should be given prior to any diagnostic procedure if there is a petechial or purpuric rash or shock.
- Septicaemic shock should be treated with appropriate volume replacement, elective intubation and ventilation.
- Patients may require inotropic support and metabolic abnormalities, coagulopathy and anemia should also be appropriately treated.
- In the presence of raised intracranial pressure it may be necessary to administer mannitol, muscle relaxation and appropriate intensive care nursing.

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- Meningococcal meningitis is a notifiable disease and there is a high risk of family members developing the disease.
 - All household close contacts should be treated to eradicate nasopharyngeal carriage.
 - It is recommended that rifampicin 600 mg 12 hourly should be given as prophylaxis for 2 days, ciprofloxacin 750 mg as a single dose is also effective.
 - Meningococcal vaccination is currently available and contains polysaccharides to serogroups A, C, Y and W135.
 - The vaccine induces protection for 3–5 years and may be of value in controlling outbreaks.

Outcome

- Complications of the acute infection are common and occur in more than 35% of patients, they include cerebral oedema, hydrocephalus, vasculitis, venous thrombosis, ventriculitis, labyrinthitis and spinal cord involvement.
- The mortality rate is approximately 25% although this rises to 40% in those with meningococcal septicaemia coexisting with meningitis.
- Of survivors, 20% have neurological sequelae such as hearing loss, loss of limbs secondary to large vessel vasculitis and neurological disability resulting from cerebral ischaemia.

Streptococcus pneumoniae (pneumococcal meningitis)

- This the most common cause of meningitis in adults over the age of 18 years (50%) with a case fatality rate of approximately 20%.
- The organism is a Gram-positive coccus; spread occurs by respiratory droplet infection.
- The primary site of colonization is the nasopharynx and the carrier state is common.
- *S. pneumoniae* meningitis is commonly caused by local extension from otitis media, or a paranasal source of infection, following a skull base fracture or sinus injury with dural tear.
- Other predisposing features include pneumonia, alcoholism, diabetes, immunodeficiency states (e.g. splenectomy, hypogammaglobulinaemia, HIV).

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- Clinical presentation is similar to other forms of pyogenic meningitis but a coexisting pneumococcal pneumonia may be present.
 - The course may be aggressive with rapid progression to coma and respiratory arrest.
 - Residual neurological sequelae are common and occur in more than 35% including cerebral edema, hydrocephalus, vasculitis, venous thrombosis, ventriculitis, labyrinthitis and spinal cord involvement.
 - Common sequelae include decreased intellectual function, memory impairment, seizures, hearing loss and dizziness, and gait disturbances.

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- The treatment of pneumococcal meningitis has been complicated by the development of penicillin and cephalosporin resistant strains.
 - Conventional treatment was with penicillin, ampicillin and ceftriaxone or cefotaxime but because of penicillin-resistant pneumococcal infection, vancomycin should be added.
 - Adjuvant dexamethasone (10 mg 6-hourly i.v. for 4 days) improves the outcome and should be commenced before or with the first dose of antibiotics.

Haemophilus influenzae

- This is a small Gram-negative coccobacillus.
- Prior to the introduction of Haemophilus influenzae type B (HiB) vaccination this was the most common cause of meningitis in children with a fatality rate of approximately 5% and permanent neurological damage in some 30% of cases.
- Meningitis occurs as a consequence of respiratory droplet spread.
- The primary treatment of Haemophilus influenzae is ceftriaxone or cefotaxime because of the significant incidence of penicillin resistant strains in up to 40%.
- Chemoprophylaxis is recommended for household contacts, 600 mg rifampicin once daily for 4 days.
- Vaccination is recommended for all children with the first dose being given at 2 months.

Listeria monocytogenes

- This is a beta haemolytic Gram-positive rod with 12 subtypes based on antigenic properties.
- The overall case fatality rate may reach 15%.
- Outbreaks are associated with contaminated food including soft cheese, unpasteurized milk and occasionally raw meat.
- Predisposing factors include pregnancy, advanced age or immunosuppression, and Listeria is particularly common in patients with malignancy, renal failure or following organ transplantation or steroid treatment.
- Listeria may cause a meningitis or meningo-encephalitis but seizures, disturbances of consciousness and movement disorders may also occur.

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- A brainstem encephalitis develops in >10%, this presents with a prodromal phase followed by progressive ponto-medullary involvement with cranial neuropathy, pyramidal and sensory signs.
 - Treatment is with Ampicillin or Amoxillin 2 g (IV) 4-hrly for 3 weeks and gentamicin for 7 days.
 - Alternatively If there is allergy to penicillins: Co-trimoxazole 3 mg/kg (IV) 6-hrly for 21 days.

Tuberculous meningitis

- Mycobacterium tuberculosis is an aerobic non-spore forming bacillus which may be cultured and stained by Ziehl–Neelsen stain.
- Infection occurs by person-to-person droplet spread.
- The organism is inhaled into the lower portion of the lungs and there multiplies locally before disseminating through lymphatic and haematological spread into other organs.
- Tuberculous meningitis develops secondary to a caseating tuberculous focus adjacent to the CSF (Rich Focus). These usually develop following prior haematogenous dissemination of mycobacteria and in the absence of pulmonary tuberculosis.

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- Tuberculous meningitis is characterized by inflammatory meningeal adhesive exudates leading to a florid small and medium vessel vasculitis which culminates in occlusion of cerebral arteries and infarction.
 - There is also a disturbance of CSF flow because of impairment of absorption at the arachnoid villi and blockage at the aqueduct and fourth ventricular outlet.
 - In tuberculous meningitis, vasculitis particularly affects the vessels at the base of the brain including the internal carotid, proximal middle cerebral and perforating vessels to the basal ganglia and internal capsule.

Clinical manifestations

- Tuberculous meningitis is often preceded by a prolonged prodrome of non-specific malaise, anorexia, low-grade fever, myalgia, photophobia and headache.
- The development of meningitis may be insidious, associated with worsening headache, nausea, vomiting, focal or generalized seizures and progressive impairment of consciousness.
- Cranial nerve palsies are common and often initially involve eye movements resulting from III, IV or VI nerve palsy.
- There may be facial weakness (VII), optic neuropathy (II), progressive hearing loss (VIII) and eventual bulbar involvement(X–XII).

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- Fundal examination may show papilloedema, optic atrophy or the presence of choroid tubercles.
 - Other neurological signs include hemiparesis and hemiplegia as a consequence of vasculitic infarction or space-occupying tuberculomas.
 - Movement disorders may be manifest as tremor, myoclonus, chorea, hemiballismus or dystonia and cerebellar ataxia may occur from direct involvement or vasculitic infarction.
 - Cerebrovascular disease usually occurs in the distribution of the anterior circulation but may be posterior in distribution.

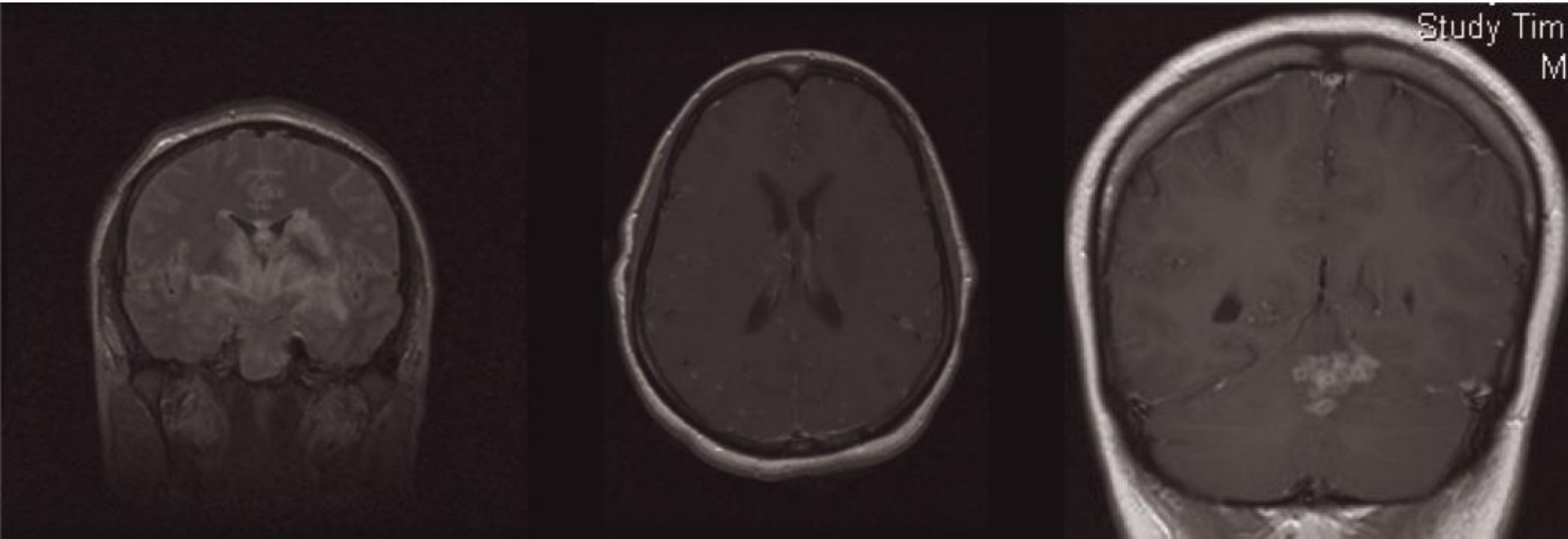
Diagnosis

- **Tuberculin skin test:** is variable in its response and unreliable.
- **CSF examination:** is characterized by the presence of **lymphocytic pleocytosis** with a high protein and low glucose.
- Direct examination of the CSF for acid–alcohol fast bacilli (AFB) requires the provision of large volumes of CSF and is experience dependent.
- AFB are only seen in up to 10% of cases with appropriate staining.
- **CSF culture:** is the diagnostic investigation but requires up to 6 weeks and is therefore of only limited value in clinical practice. Culture is positive in up to 10-60% of cases.
- Nucleic acid amplification techniques (NAATs) for the diagnosis of TBM are 98% specific but only 56% sensitive.
- **CSF PCR:** The use of real- time PCR can improve the sensitivity but is expensive.

Radiology

- Approximately 50% of patients with tuberculous meningitis may show evidence of previous tuberculosis on **chest X-ray** with up to 10% having miliary tuberculosis.
- **CT brain** is commonly abnormal with marked enhancing exudate in the basal cisterns.
- There may also be hydrocephalus, parenchymal enhancement, evidence of cerebral infarction or cerebral oedema or focal tuberculoma.
- **MRI** is sensitive in showing meningeal enhancement, focal parenchymal abnormalities or the development of communicating or obstructive hydrocephalus.

Tuberculous meningitis



Management of tuberculous meningitis

- Treatment should commence with **isoniazid**, **rifampicin**, **pyrazinamide** and **ethambutol** for 2 months and this should be followed by a 9-12 month continuation phase of treatment with isoniazid and rifampicin.
- All antituberculous treatment carries significant toxicity.
- Steroids are indicated in the acute phase particularly if there is cerebral oedema, spinal block, severe tuberculous meningitis, spinal arachnoiditis or cerebral vasculitis.
- Surgical intervention may be required in the presence of obstructive hydrocephalus that has not responded to medical treatment.
- The outcome of tuberculous meningitis has remained poor and the mortality is still >20%, with severe neurological sequelae occurring in up to 30% of survivors.

Treatment of tuberculous meningitis

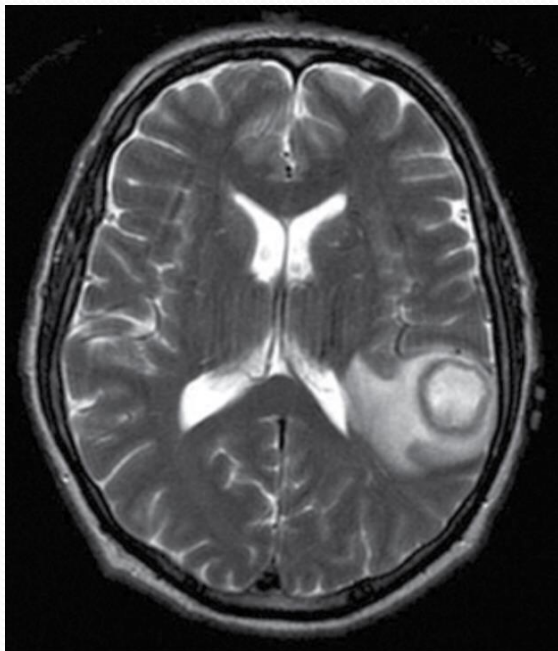
	Adult daily dose	Comment	Side effects
Isoniazid	300 mg	Essential drug Penetrates CSF freely and has potent early bacteriocidal activity Add pyridoxine to avoid peripheral neuropathy	Hepatitis, haemolytic anaemia, aplastic anaemia, peripheral neuropathy, optic neuritis, mania, fits
Rifampicin	450 mg (wt <50 kg) 600 mg (wt >50 kg)	Essential drug Bacteriocidal but poor CSF penetration (10%)	Hepatitis, thrombocytopenia, headache, confusion, drowsiness
Pyrazinamide	1.5 g (wt <50 kg) 2.0 g (wt >50 kg)	Bacteriocidal with good penetration	Hepatitis, anorexia, flushing
Fourth drug: use one of			
Streptomycin	20–40 mg/kg	Neither penetrates CSF well in absence of inflammation (10–50%)	Avoid in pregnancy and renal impairment Neuromuscular blockade
Ethambutol	15 mg/kg		Retrobulbar neuritis, peripheral neuropathy, confusion
Ethionamide	500–750 mg	Penetrate healthy and inflamed meninges but no evidence of advantage over other drugs	

Cerebral abscess

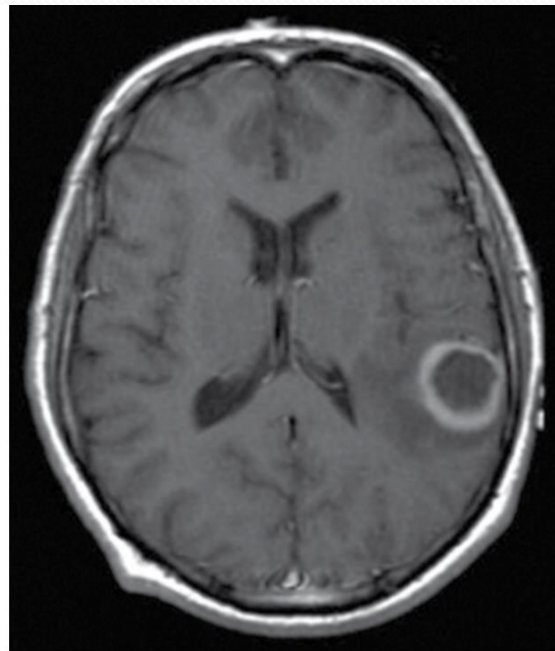
- This is a focal suppurative (pus forming) infection occurring within the cerebral parenchyma.
- It develops as a result of contiguous spread of infection from paranasal sinuses, mastoiditis, otitis media, osteomyelitis or following postoperative and posttraumatic infections.
- Less commonly, it may arise as a result of haematogenous spread from distant sites including teeth and lungs.
- The causative agents depend on the underlying abnormality, and the age and immunological status of the patient.
- Cerebral abscesses that arise from dental, frontal or ethmoid sinuses tend to involve the frontal lobe while those arising from the sphenoid sinuses or otitic infection particularly involve the temporal lobes.

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- Cerebral abscesses resulting from haematogenous spread tend to occur as multiple rather than single abscesses , often in the region of greatest blood flow, within the basal ganglia.
 - The onset is with subacute or indolent fever, headache, nausea, vomiting, seizures or focal neurological signs.
 - Fever is variable and may be absent in up to 50% of patients usually those who are older; it tends to resolve rapidly with steroids.
 - Rupture of an abscess into the ventricles may manifest as a severe headache and developing signs of meningism.

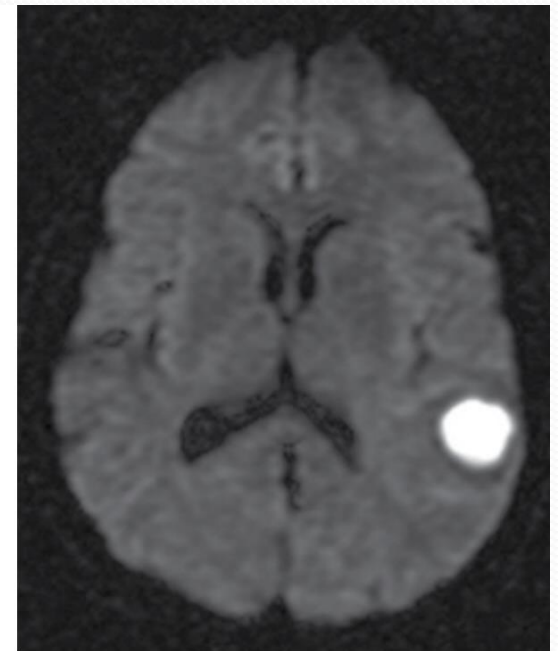
Left temporal abscess MRI: (a) T2-weighted, (b) T1 weighted post-contrast and (c) diffusion weighted images.



(a)



(b)



(c)

CT scan showing multiple bacterial abscesses in a 53-year-old woman with Gram-negative septicaemia.



Investigation and Treatment

- Routine hematology and biochemistry usually shows an elevation of the erythrocyte sedimentation rate (ESR), white cell count and blood cultures are only positive in about 10%.
- CT scan may show a thin enhancing ring of uniform thickness.
- MRI with enhancement is more sensitive and will document the extent of surrounding edema more accurately.
- Lumbar puncture is contraindicated in patients with mass lesions.
- Treatment is urgent, necessitating administration of appropriate antibiotic: third or fourth-generation cephalosporin (e.g., cefotaxime, ceftriaxone, or cefepime) and metronidazole.
- In patients with penetrating head trauma or recent neurosurgical procedures, treatment should include ceftazidime as the third-generation cephalosporin to enhance coverage of *Pseudomonas* spp. and vancomycin for coverage of staphylococci.
- Meropenem plus vancomycin also provides good coverage in this setting.

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- Aspiration and drainage of the abscess under stereotactic guidance are beneficial for both diagnosis and therapy.
 - Complete excision of a bacterial abscess via craniotomy or craniectomy is generally reserved for multiloculated abscesses or those in which stereotactic aspiration is unsuccessful.
 - Medical therapy alone is given for patients whose abscesses are neurosurgically inaccessible, for patients with small (<2–3 cm) or nonencapsulated abscesses (cerebritis), and patients whose condition is too tenuous to allow performance of a neurosurgical procedure.
 - All patients should receive a minimum of 6–8 weeks of parenteral antibiotic therapy.
 - patients should receive prophylactic anticonvulsant therapy because of the high risk (~35%) of focal or generalized seizures.
 - Anticonvulsant therapy is continued for at least 3 months after resolution of the abscess

Tetanus

- Tetanus is a disease preventable by immunization caused by the neurotoxin tetanospasmin, elaborated by the anaerobic Gram-positive rod *Clostridium tetani*.
- It is found in soil and feces which persists as resilient spores capable of surviving many years and resistant to most disinfectants and boiling for 20 minutes.
- The spores germinate under appropriate anaerobic conditions.
- The toxin is produced in the wound and binds to peripheral motor nerve terminals before being transported via retrograde axonal transport into the spinal cord and/or brainstem.

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- The toxin ultimately migrates to the presynaptic terminals inhibiting release of gamma-aminobutyric acid (GABA) and glycine, important inhibitory neurotransmitters.
 - In the absence of the inhibitory influence of GABA and glycine, alpha motor neurons fire rapidly, producing muscle spasms and rigidity.
 - Preganglionic sympathetic neurons are also affected, resulting in increased catecholamine levels and sympathetic over-activity.

Epidemiology

- Tetanus occurs in neonates because of contamination of the Umbilical stump.
- Children and adults may develop tetanus as a consequence of infected wounds, skin lacerations, as a consequence of intramuscular injections or drug abuse and occasionally after abdominal surgery.
- In adults incidence is greatest at ages >65 years, usually as a result of inadequate reimmunization and waning immunity.

Clinical features

- The incubation period varies from a few days to several weeks.
- There may be localized spasm and rigidity in the region of the wound.
- The onset may be with back pain, increased muscle tone and rigidity of the masseter muscles leading to trismus (**lockjaw**) and a similar rigidity of the facial muscles (**risus sardonicus**).
- There is a localized stiffness near to the injury with sustained rigidity of the axial muscles with involvement of the neck, back and abdomen and, in severe cases, reflex spasms and opisthotonus.
- Paroxysmal contractions of muscles appear in response to slight stimuli, and in severe cases are bad enough to cause limb fractures, tendon avulsion and rhabdomyolysis.

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- Spasms of respiratory muscles lead to asphyxia, vocal cord obstruction and aspiration associated with increased bronchial secretions, hypersalivation and dysphagia.
 - Autonomic manifestations are common in severe tetanus with profuse sweating, hypersalivation and extreme hyperpyrexia.
 - Fluctuations of blood pressure and heart rate are cardinal features and these may be followed by arrhythmias and circulatory failure.
 - There may be transient glycosuria.
 - There is also excessive bronchial secretion, gastric stasis, diarrhea, acute renal failure and volume depletion.

Diagnosis

- The diagnosis is predominantly clinical, from the history of the spasms and examination findings.
- Culture of *C. tetani* from a wound is confirmatory.
- Serology can also have a supportive role in diagnosis, with serum tests for tetanus toxin and tetanus toxin antibody.

Treatment

- Treatment in the ICU has resulted in a marked improvement in prognosis for patients with tetanus but the mortality is still approximately 10%.
- Severe muscular rigidity may last for weeks and assisted ventilation may be required for several weeks.
- Complete recovery is typical, although mild painful spasms can persist for months.

Neutralise any unbound toxin	TIG intramuscular in multiple sites at 5000–10 000 IU. In the event of reduced availability of TIG, intravenous human normal immunoglobulin can also be used
Debridement of the wound	This encourages eradication of tetanus bacteria and arrest of toxin release
Antibiotics	Metronidazole 500 mg 4 times daily for 7–10 days
Supportive therapy during the acute phase	Nursing in a calm, quiet environment with cardiorespiratory monitoring Fluid balance, nutrition and the control of muscle spasms with diazepam, midazolam Tracheal intubation, ventilation and neuromuscular blockade with vecuronium
Immunisation	Immunisation is recommended following recovery from tetanus Primary childhood immunisation programmes as five doses of combined DPT in the UK given at 2 months, 4 months, 6 months, 3–5 years and 13–18 years Boosters are recommended every 10 years



Viral disease of the nervous system

Viral meningitis

- Viruses cause an isolated aseptic meningitis characterized by symptoms and signs of meningeal irritation with a CSF pleocytosis.
- **Non-polio enteroviruses** are the most common cause of viral meningitis, these include coxsackie and echovirus strains.
- The main causes include:
 1. Echovirus
 2. Coxsackie A, B
 3. Enterovirus 70, 71
 4. Mumps
 5. Measles
 6. Herpes simple virus 2 (HSV-2)
 7. Varicella zoster virus (VZV)
 8. Epstein–Barr virus (EBV)
 9. Cytomegalovirus (CMV)

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- Viral meningitis is a relatively uncommon complication of systemic viral infection.
 - Most viruses gain access to the body from the oropharynx. They are amplified (multiply) in lymphatic tissue, spread to the bloodstream (viraemia) and cross the choroid plexus or capillary endothelial cells to reach the CNS.
 - In viral meningitis there may be a flu-like prodrome followed by the sudden onset of intense frontal headache, fever and neck stiffness associated with photophobia, malaise, myalgia and severe nausea and vomiting.
 - Although there is pyrexia, neck stiffness and meningeal signs, patients are generally less unwell than those with bacterial meningitis.

Diagnosis

- The peripheral white cell count is usually normal but may be increased or decreased and liver function may be abnormal.
- The CSF is clear and colorless with a normal to moderately elevated pressure.
- The cell count may be up to 1000 cells/mm³, the majority are lymphocytes.
- Viral isolation is undertaken from the throat, urine or stool and antibody studies are possible in serum or CSF.
- The detection of viral RNA or DNA is now undertaken using PCR in the serum or CSF.

Management

- Treatment is supportive but it is usual to admit a patient to exclude bacterial meningitis.
- A clinical diagnosis of viral meningitis is not secure; substantial errors have occurred by presuming a viral infection, when the underlying infection has been pyogenic or tuberculous.
- Herpes virus meningitis should be treated with **aciclovir** (**famciclovir**, **valaciclovir**, **ganciclovir** and **foscarnet** are alternatives).
- The only therapy of clinical use for enterovirus meningitis is **immune serum globulin**.
- The prognosis in viral meningitis is good with spontaneous recovery usually occurring within 1–2 weeks.

Encephalitis

- Encephalitis is acute infection of the parenchyma of the brain caused by a virus, which results in a diffuse inflammatory process, often also involving the meninges.
- Encephalitis causes focal or multi-focal neurological deficits or seizure activity.
- the most common severe forms of infectious encephalitis are those caused by:
 1. HSV-1
 2. VZV
 3. EBV
 4. CMV
 5. Enteroviruses
 6. Adenoviruses
 7. Influenza virus A and B

Clinical features

- The clinical manifestations of encephalitis are often most severe in infants and those over the age of 65.
- It is characterized by fever in 90% of patients.
- Seizures and impaired level of consciousness are common and help to distinguish acute infective encephalitis from viral meningitis.
- The presence of headache, pyrexia and meningism suggest leptomenigeal irritation while parenchymatous involvement leads to focal neurological signs including seizures and alteration of consciousness progressing to stupor and coma.
- More commonly, behavioral and speech disturbances develop and abnormal movements are associated with lesions in the basal ganglia.

Herpes simplex encephalitis

- The majority of cases of herpes simplex encephalitis (HSE) in adults are caused by **HSV-1**.
- Primary infection usually develops in the oropharyngeal mucosa before the virus is transported by retrograde transneuronal spread via the trigeminal nerve to establish latency in the olfactory bulb or trigeminal ganglion.
- The presence of labial herpes has no diagnostic specificity to HSE.
- The herpes virus leads to inflammation, infection and necrotizing lesions particularly in the inferior and mesial temporal lobes which may also involve the orbital frontal cortex and limbic structures.

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- The onset of HSE is with fever, headache and alteration of consciousness which may develop gradually or rapidly over a matter of hours.
 - The most common manifestations are personality change, dysphasia with progressive behavioral disturbance and occasional psychotic features.
 - Less typical features include the development of hemiparesis or a visual field defect (particularly superior quadrantic).
 - Focal or generalized seizures are often associated with olfactory or gustatory hallucinations.

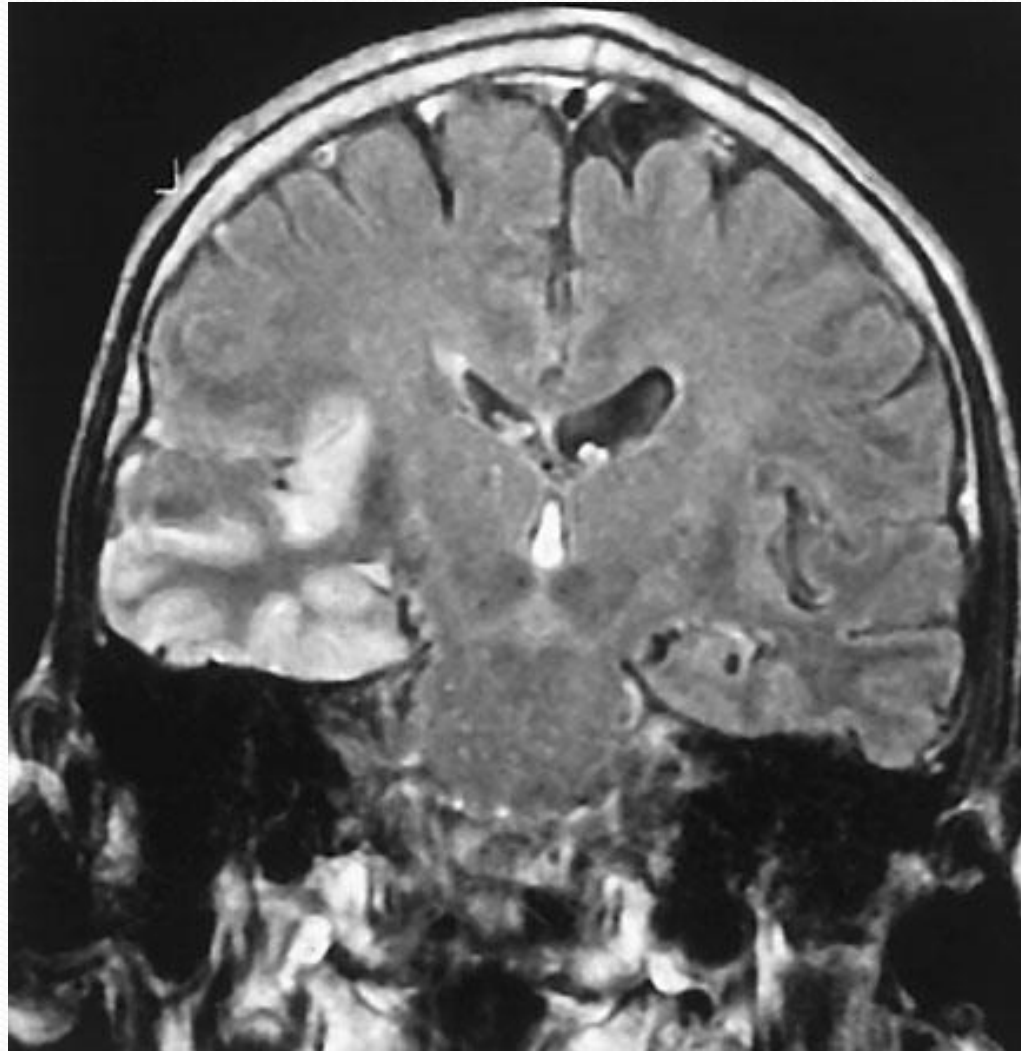
Investigation

- The **MRI** characteristically shows high signal areas of unilateral or bilateral focal edema on T2 images in the medial and inferior temporal lobes.
- The electroencephalogram (**EEG**) is characterized by periodic stereotyped, lateralized sharp and slow wave complexes occurring at regular intervals between 2 and 3 seconds.
- **CSF** is generally under an increased opening pressure and may show a mild to moderate lymphocytic pleocytosis of 5–500/mm³; there may be a mild to moderate elevation in the protein (0.6–6 g/L) and a normal or mildly decreased glucose.
- **PCR** examination of the CSF is sensitive and specific for the detection of HSV DNA.
- **Brain biopsy** is now rarely undertaken but may be considered if diagnostic uncertainty remains.

PLEDs



MRI from a patient with herpes simplex encephalitis



Management

- High-dose intravenous aciclovir (10–15 mg/kg body weight 8 hourly) reduces the mortality of HSE from 70% to 20%.
- It is also appropriate treatment for VZV-related CNS disease.
- Treatment should be continued for 14–21 days or at least until the PCR has become negative.
- Occasionally, HSV may be resistant to aciclovir and foscarnet is indicated in this situation.
- Patients with HSE can develop seizures, which require specific treatment
- Corticosteroids may be of value if cerebral edema develops and, occasionally, intracranial pressure monitoring and even surgical decompression may be necessary.

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- Although high-dose aciclovir has reduced mortality from HSE, there is still a significant morbidity that may occur including severe memory impairment (up to 69%), personality behavioral changes (up to 45%), epilepsy (up to 25%) and aphasia (up to 4%).
 - Early treatment is essential for the reduction of morbidity, particularly before the development of impairment of consciousness.

Rabies

- It is caused by a lyssavirus which is inoculated into the tissue of a wound usually caused by a canine or bat bite.
- The virus may replicate locally in muscle cells or attach directly to nerve endings.
- It then enters the presynaptic nerve terminal and is carried in a retrograde direction.
- On reaching the CNS there is massive viral replication in neurons before transynaptic transmission of the virus occurs from cell to cell.
- Viral proteins accumulate in the cytoplasm appearing as inclusions (**Negri bodies**).

Clinical features

- Clinical features develop 1–3 months after the bite.
- The onset is with a prodrome of fever, nausea, vomiting, chills, malaise, fatigue and insomnia together with the development of pain, pruritus, paranesthesia and fasciculation close to the bite site.
- This is followed by the development of progressive encephalopathy.
- 1. The disease is described as either **furious rabies** (80%) or **paralytic rabies** (20%).
- **Furious rabies** is characterized by fluctuating episodes of excitement and hyper-excitability with periods of lucidity.
- Aggressive behavior, confusion and hallucinations develop.
- Autonomic dysfunction is manifest as hypersalivation, sweating and pilo-erection.

- Patients develop severe muscle spasms of the throat, trunk and respiratory muscles triggered either by the sight or sound of water or attempts to drink (**hydrophobia**) or even by a draft of air (**aerophobia**).
- The spasms cause generalised extension, convulsions and opisthotonos with death usually occurring within a week of the onset of furious rabies as a consequence of cardiac arrhythmias, myocarditis and respiratory arrest.
- **Paralytic rabies** is associated with an ascending paralysis leading to constipation, urinary retention, respiratory failure and an inability to swallow.

Diagnosis

- The clinical features may be obvious.
- A CSF mononuclear pleocytosis develops, usually within the first week, but serum neutralizing antibodies are not present until the tenth day of the illness.
- Viral isolation is occasionally possible from saliva, throat, trachea, CSF or brain biopsy and skin biopsy and immunofluorescent antibody techniques will demonstrate an antigen in the small nerves of the skin taken from the nape of the neck, often in a hair follicle.
- PCR detection of rabies virus RNA has also been demonstrated from the brain, saliva and CSF.
- Postmortem virus isolation and brain cultures should be taken if possible.

Management

- Management is supportive.
- Rabies is almost invariably fatal.
- Patients with rabies need heavy sedation and adequate analgesia to relieve terror or pain.
- Acute interventional care may be necessary to manage cardiac arrhythmias, cardiac and respiratory failure, raised intracranial pressure, convulsions and fluid and electrolyte disturbances.
- Only very occasional survival has been documented, usually in patients who had received rabies vaccine before the onset of symptoms.
- Immunization before possible exposure is strongly recommended to travelers who plan to visit regions where rabies is endemic, as it provides protection after unrecognized exposure and may simplify treatment.

Post-exposure treatment of rabies

Post-exposure prophylaxis

If there is any doubt about the possibility of rabies then post-exposure prophylaxis should begin immediately

- **Local wound care** – washed thoroughly with soap and water, debridement as necessary and tetanus prophylaxis
- **Active immunisation** – using either human diploid cell vaccine or alternatively purified chick embryo cell culture vaccine in five doses given over 28 days. If a bite occurs in a previously immunised individual only two booster doses are necessary
- **Passive immunisation** – human rabies immunoglobulin should be given immediately before the vaccine. The wound should be infiltrated with the immunoglobulin and then a dose given intramuscularly

Examination of the biting animal

Ideally, the animal should be captured confined and observed for 10 days
If there is any clinical suspicion of rabies, the animal should be killed and the brain examined for virus

Antigen detection in animals

Using fluorescent antibody techniques or cell culture of mouse inoculation